

Modeling Football Player Trajectories During Passes Using Graph-Structured Recurrent Networks

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Abstract

Accurately **forecasting player trajectories** during the ball-in-flight phase of American football requires models that capture **multi-agent interactions** while remaining consistent with underlying physical constraints. This work evaluates how **input normalization and target parameterization** influence the performance of a Spatio-Temporal Graph LSTM model. Three normalization strategies are considered, along with two prediction targets, and the results show that **RMS scaling combined with velocity prediction provides the highest accuracy**, with an RMSE of 0.684. The findings indicate that normalization choices have a major effect on forecasting quality and that carefully selecting both the input scaling method and the target representation is essential for reliable trajectory prediction.

Introduction

- Downfield passing plays create high variability in player movement and lead to some of the most consequential events in American football.
- 2026 NFL Big Data Bowl Prediction task: from tracking data, produce a sequence of predicted (x, y) coordinates for each relevant player until the ball lands.

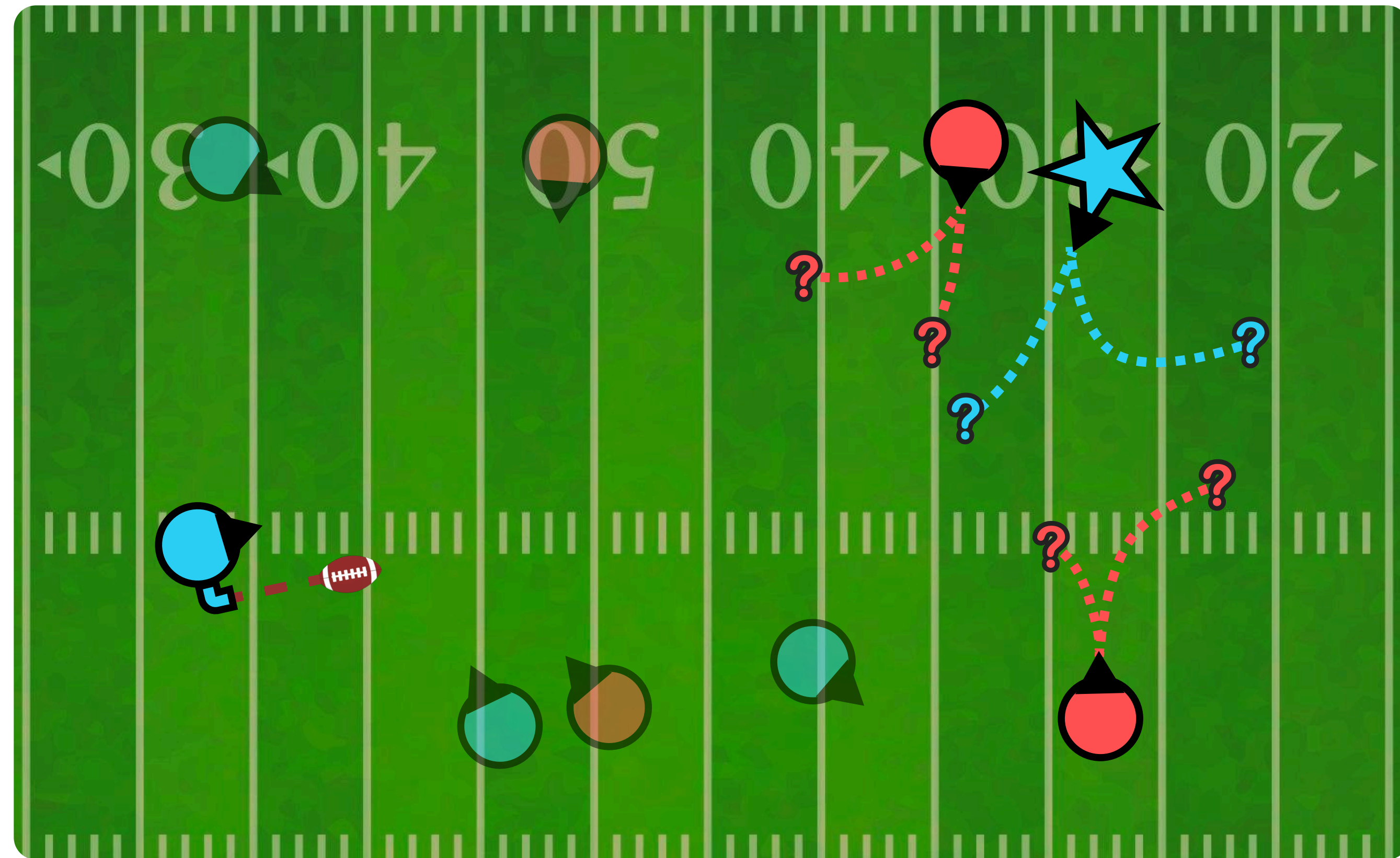


Figure 1. Illustration of the forecasting task. Player positions are observed until ball release, after which a model predicts player trajectories while the ball is in the air. Only the targeted receiver, symbolized with a star, and relevant defenders have their trajectories scored.

Feature Type	Examples
Identity	Game, play, and player identifiers for grouping sequences.
Player	Position, physical measurements, team side, and role in the play.
Spatial	Absolute field coordinates, orientation, and directional movement.
Kinematic	Instantaneous speed and acceleration at each frame.
Contextual	Play direction, situational variables, and ball landing location.

Table 1. Summary of the major feature categories used in the trajectory forecasting model, including identifiers for sequence organization, player attributes, spatial location, kinematic measurements, and broader situational context.

Methodology

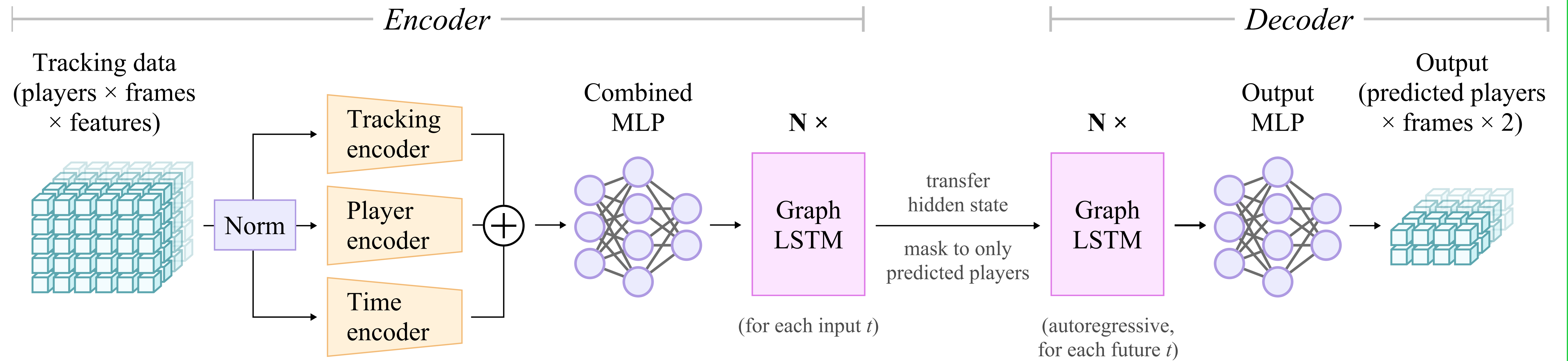


Figure 2. High-level architecture of the Spatio-Temporal Graph LSTM model. Input tracking data is normalized, encoded with tracking, player, and time encoders, fused through an MLP, and processed by stacked Graph LSTM layers. Decoder states are initialized from the encoder and generate future positions autoregressively.

Graph Attention + Long Short-term Memory (spatial dependencies) (temporal dependencies)

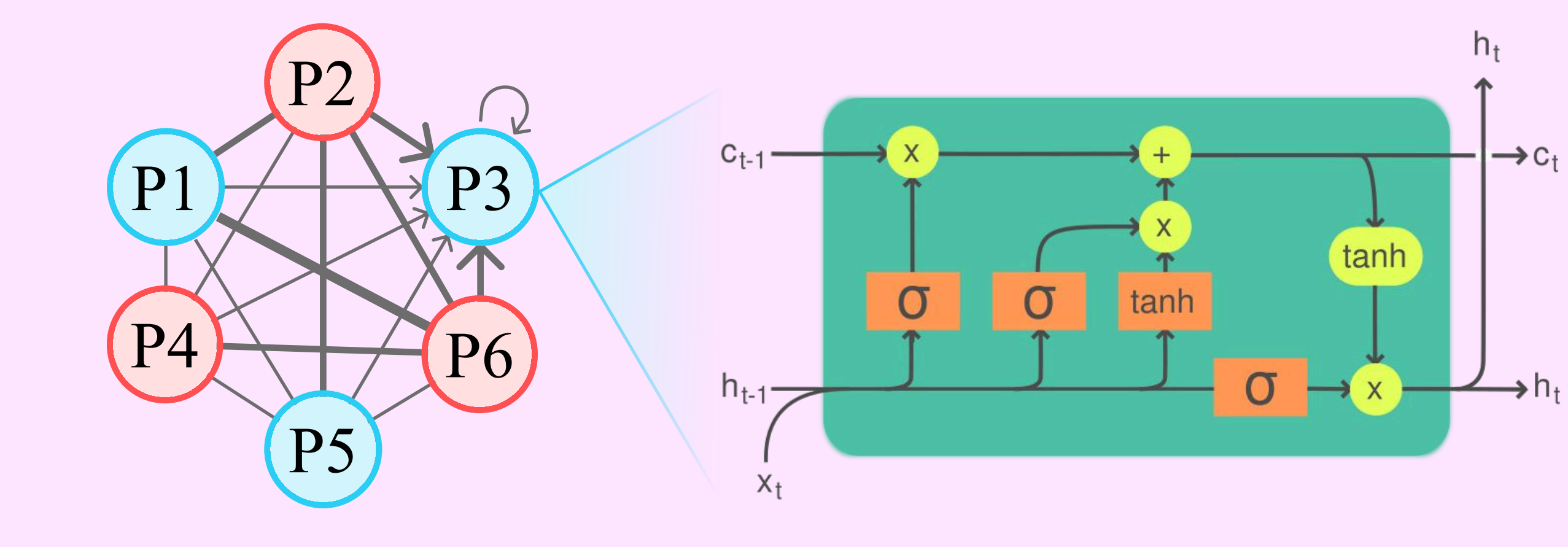


Figure 3. Illustration of the Graph Attention and LSTM components within each Graph LSTM block. The attention module models spatial interactions among players, while the LSTM cell captures temporal dependencies across frames.

Experiments

Normalization strategies

- Field scaling:** Positions are divided by the field size to put them in a 0-to-1 range. Other features like speed and acceleration are left as-is.
- Max scaling:** Features are divided by their known maximum values. Positions still use field scaling.
- RMS scaling:** Positions are centered at the field midpoint and divided by half the field size, putting them in a -1 to 1 range. All other features (speeds, accelerations, distances) are divided by the root-mean-square value of those features.

Prediction targets

- Velocity**
- Acceleration**

Setup

- 10GB VRAM
- 10 CPU cores
- 400 epochs
- 16 epoch patience for early stopping
- N = 2 Graph LSTM layers

Results

Normalization	Target Variable	RMSE ↓
Field	Velocity	0.755
Field	Acceleration	0.936
Max	Velocity	0.702
Max	Acceleration	0.911
RMS	Velocity	0.684
RMS	Acceleration	0.699

Table 2: Test root-mean-square error (RMSE) for Graph LSTM trajectory forecasts under six configurations that combine three input-normalization methods with two target formulations. Errors represent average positional deviation in yards for all predicted post-release frames.

- RMS scaling across all features achieves the lowest errors.
- Simple field scaling is less effective than balanced feature normalization.
- Velocity as a prediction target outperforms acceleration.

Discussion and Conclusion

- RMS scaling preserves physical relationships while reducing feature-scale disparities, making it particularly effective for graph-recurrent networks.
- Acceleration-based targets introduce additional uncertainty and integration error, limiting predictive accuracy.
- Graph LSTM effectively leverages multi-agent interactions, showing the value of relational modeling in trajectory forecasting.

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